Assessment 4 – Research Poster

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1. Research Focus

• Final research question(s):

Does delivering interventions at a machine learning (ML)-detected time lead to higher receptivity than delivering interventions at a random time?

• Refined research scope

Research shows that psychological interventions after successful cancer treatment can be sources of stress. Using AI to predict when it's the best time approach people has been explored in other domains, but not in psycho-oncology.

3. Research Design Overview

• Methodology and research design:

Use Deep Learning techniques and EMA data to predict the best time to approach subjects that recently finished their cancer treatment.

• Data collection and analysis approach:

Preprocess sensor data and EMA data from cancer survivors to assess receptivity to mental health interventions.

4. Project To-Do List and Timeline

• Remaining tasks

All associated to content for chapters:

- Methodology
- Results and analysis
- Conclusions

• Indicative timeline to completion

Dissertation	07/26	08/26	09/26	10/26	11/26	12/26	01/27	02/27	03/27	04/27	05/27	06/27	07/27	08/27
Data exploration														
Create data preprocessing pipeline														
Submit final literature review														
Model selection														
Hyperparameter space exploration														
Formalise methodology														
Performance and quality comparison														
Results gathering														
Analysis														
Conclusions														
Cushion														

2. Literature Review Structure

• Key themes or sections

- 2.1 Sensing mental health
  - 2.1.1 Mobile health
  - 2.1.2 Active digital phenotyping
    - 2.1.2.1 Ecological Momentary Assessments
    - 2.1.2.2 Interventions
  - 2.1.3 Passive digital phenotyping
- 2.2 Machine learning in healthcare
  - 2.2.1 Sensor inference
  - 2.2.2 Ethical considerations
- 2.3 State of the art
  - 2.3.1 Predicting receptivity
  - 2.3.2 Time series classification

• Core sources underpinning each theme

- 2.1 (Verma et al. 2024)
  - 2.1.1 (Verma et al. 2024)
  - 2.1.2 (Verma et al. 2024)
    - 2.1.2.1 (Künzler et al. 2019)
    - 2.1.2.2 (Dao et al. 2021)
    - 2.1.3 (Mishra et al. 2021), (King et al. 2024)
- 2.2 (Rani et al. 2025)
  - 2.2.1 (Sarker et al. 2014), (Katevas et al. 2017)
  - 2.2.2 (Kazim, 2021)
- 2.3 (Heathcote et al. 2022)
  - 2.3.1 (Künzler et al. 2019), (Mishra et al. 2021), (King et al. 2024)
- 2.3.2 (Wang et al. 2025)

• Identified research gap

The availability for interventions of cancer survivors will change on a person-to-person basis. Therefore, there is no consensus on what the best time is for AI interventions for high-patient receptivity is.

5. Iteration and Learning

• Key changes made during the module:

1) Enriched more my reference framework.
2) Improved the description of the dataset at hand.
3) Specified better what techniques for 'mood measurement' have been already implemented.
4) Explain more the data preprocessing involved.

• Why these changes were made

For 1) To have better definitions when required.
For 2) Strengthen methodology justification and show better how deep learning is a respectable approach to deal with the data.
For 3) To justify better the research gap and why this project will be useful.
For 4) Initially I didn't have enough time to develop this.

• What was learned from the iteration process

In general, the right structure for chapters 1 and 2, the foundation of the dissertation.

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